

SquidWork Database Project

Phase 1 Progress Submission - Conceptual Design

This document reports Phase 1 only. Logical mapping, SQL implementation, mock data, UI/backend integration, and final defense are future milestones.

A. Project Identity

Item	Information
Course	INT1313 - Databases
Project	#3 - Freelance Marketplace Ecosystem (Mini-Upwork)
Team	Squid
Submission	Phase 1 - Weeks 3-4 conceptual design
Status	In progress; Phase 1 design prepared

Team members and current ownership

Member	Email	Responsibility
Le Minh Thien	n24dccn167@student.ptithcm.edu.vn	Requirements, business rules, EER design
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Nguyen Tuan Vy	n24dccn184@student.ptithcm.edu.vn	Scenario analysis and relationship review

1. Scope and problem statement

SquidWork is a small freelance marketplace database concept. It records how a client posts a job, how freelancers submit proposals, how an accepted proposal becomes a contract, and how the contract is divided into milestones. Reviews are included because reputation is part of how users choose to work with one another.

The Phase 1 boundary is conceptual: identify the users, business objects, rules, and relationships needed by the scenario. Implementation details are deferred to later phases.

2. Requirements

2.1 Functional requirements

ID	The system must...
FR-01	register a user and store common account information.
FR-02	allow a user to act as a client, a freelancer, or both.
FR-03	allow a client to post and manage jobs.

ID	The system must...
FR-04	allow a freelancer to submit a proposal for a job.
FR-05	allow an accepted proposal to create one contract.
FR-06	allow a contract to contain ordered milestones.
FR-07	record payments and transactions related to milestones.
FR-08	allow parties to review each other after a contract.
FR-09	summarize ratings received by a user.

2.2 Non-functional requirements

ID	Requirement
NFR-01	Preserve referential integrity between related business objects.
NFR-02	Use consistent identifiers and names suitable for relational implementation.
NFR-03	Trace each business rule to an entity or relationship.
NFR-04	Keep the core marketplace lifecycle extensible.

3. Business rules

ID	Business rule
BR-01	Each user has a unique identifier and email address.
BR-02	A user may be a client, a freelancer, or both.
BR-03	A client can post many jobs; each job belongs to one client.
BR-04	A job belongs to one category and can receive many proposals.
BR-05	A freelancer can submit at most one proposal for a specific job.
BR-06	An accepted proposal may create at most one contract.
BR-07	Each contract has one client, one freelancer, and many milestones.
BR-08	Milestones are ordered within a contract.
BR-09	A milestone may have zero or one payment.
BR-10	A review belongs to one contract and has one reviewer and one reviewee.
BR-11	A reviewer cannot review the same reviewee twice for the same contract.
BR-12	A user can have one rating summary derived from reviews received.

4. EER model

USER is the supertype. CLIENT_PROFILE and FREELANCER_PROFILE are overlapping subtypes because the same account may use both roles. FREELANCER_SKILL resolves the freelancer-skill many-to-many relationship in the later logical design.

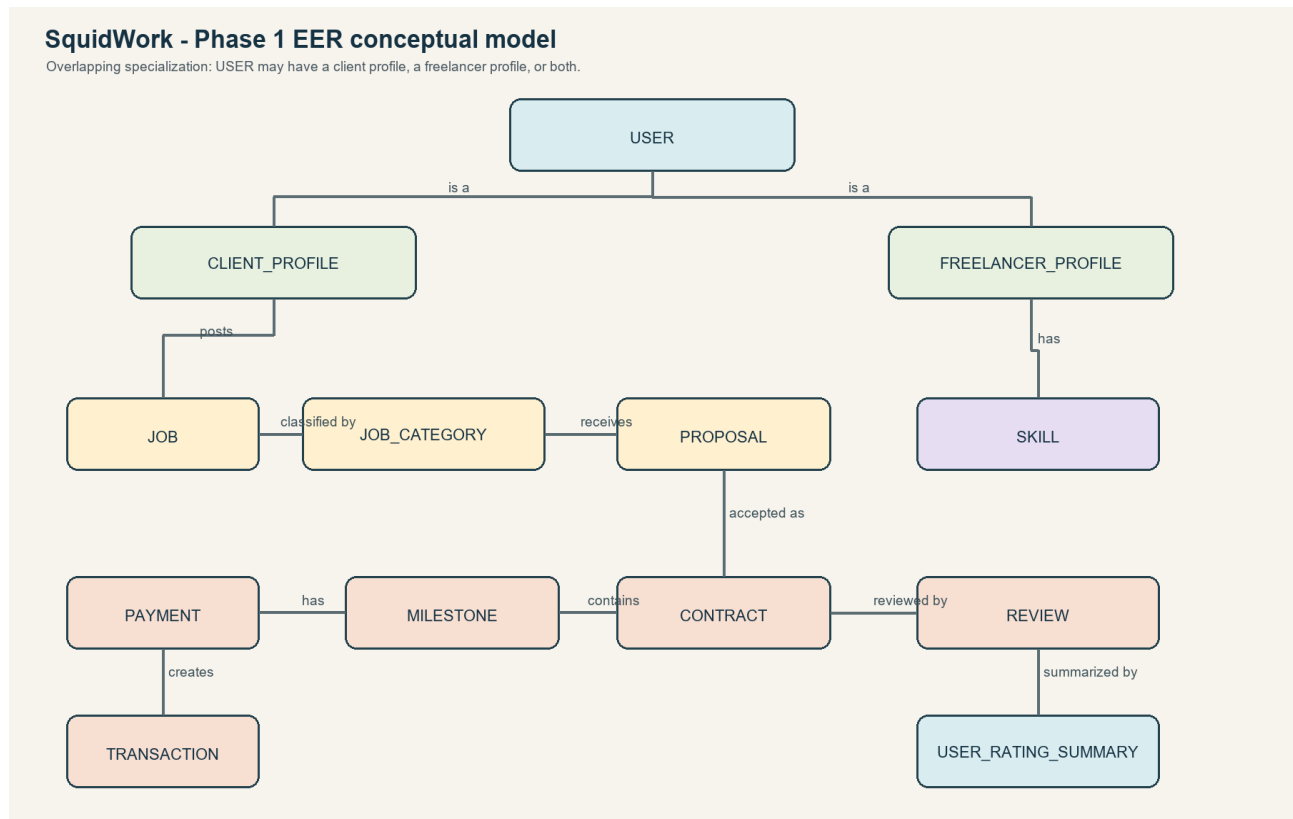


Figure 1. SquidWork Phase 1 EER conceptual overview

Relationships and cardinalities

Relationship	Cardinality	Explanation
USER - CLIENT_PROFILE	1 to 0..1	A user may have zero or one client profile.
USER - FREELANCER_PROFILE	1 to 0..1	A user may have zero or one freelancer profile.
CLIENT_PROFILE - JOB	1 to N	A client may post many jobs.
JOB - PROPOSAL	1 to N	A job may receive many proposals.
FREELANCER_PROFILE - SKILL	M to N	Resolved through FREELANCER_SKILL.
PROPOSAL - CONTRACT	1 to 0..1	Only an accepted proposal may create a contract.
CONTRACT - MILESTONE	1 to N	A contract contains ordered milestones.
MILESTONE - PAYMENT	1 to 0..1	A milestone may be unpaid or have one payment.
CONTRACT - REVIEW	1 to N	A completed contract may receive reviews.

5. Status and next steps

Completed for this submission: scenario analysis, scope boundary, requirements, explicit business rules, overlapping specialization, entity list, and relationship/cardinality analysis.

Next: relational mapping and normalization, then DDL/mock data/queries/views/triggers, and finally UI/backend integration. These items are not claimed as completed in this Phase 1 report.